

Dealing with a Differentiated Whole: The Philosophy of the WSR Approach

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This paper seeks to explore a philosophical ground for the formulation and practice of an Oriental systems methodology, the *wuli-shili-renli* approach (WSR). Drawing insights from ancient Chinese thought, WSR contends that, in sociotechnical systems design/management, we should investigate and follow *wuli*, *shili*, and *renli*, which constitute a differentiable, dynamic whole. *Wuli*, *shili*, and *renli* condition and determine the fate of our projects, meaning different methods are needed to deal with each differentiable aspect. It also points out that, to deal with *wuli*, *shili*, and *renli* properly, we need to engage in encounters with the “dark side” of Confucianism and with the contemporary tendency of scientism.

KEY WORDS: systems methodology; methodological pluralism; Confucianism; Chinese philosophy; systems philosophy; East and West; comparative systems studies; *wuli*, *shili*, *renli*; systems engineering.

1. INTRODUCTION

Encounters between modern Western sciences and ancient Eastern philosophies figure among the most fruitful features of this century and, I believe, it will also be so in the centuries to come. This appears to be particularly true in the field of systems thinking and systems science. During the last couple of years, we have witnessed within the international systems community persistent efforts to explore possibilities and opportunities for cross-cultural learning and integration (see, e.g., Zukav, 1979; Macy, 1991; Pressman, 1992; Capra, 1992; Mandel, 1995). Basically, these efforts focus on investigating insights from the Eastern philosophies; illustrating the value of those insights; and demonstrating the unrecognized and surprising similarities and compatibility between modern systems science, which was initiated mainly in the West, and Eastern philosophies—especially the ancient ones.

However, in this development, one aspect or direction has not been touched.

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That is, how to reconstruct and transform those insights into practical (i.e., operational) approaches, so that the values of Eastern culture can more easily be incorporated into systems research and practice, rather than remaining merely pedagogical.

This paper tries to seek a philosophical ground for the formulation and practice of an Oriental systems methodology, the *wuli–shili–renli* approach (WSR), first proposed by Gu and Zhu (1995), which I consider to be an effort to fill this gap.

To Gu and Zhu, WSR is Oriental because, first of all, it is inspired by, and based on, the main current of Chinese cultures.² It is Oriental also because it is formulated upon the real experiences of systems practice in the Oriental context since the 1950s.³ Further, it is Oriental because our original purpose of developing this approach was, and largely remains, to improve and popularize systems thinking in the Oriental context in a more locally meaningful and thereby convincing manner.

Needless to say, WSR is by no means purely Oriental. I do not believe that there can be anything “pure” any longer, in the sense of being uninfluenced; i.e., not benefiting from, or polluted by (depending on one’s viewpoint), developments from “outside.” Rather, in our formulation of WSR, we have consciously and insistently tried to incorporate and integrate, as much as possible, the latest achievements from the systems movement worldwide, especially those of Churchman (1968a, b, 1971, 1979), Flood and Jackson (1991a, b), Habermas (1971, 1981a, b), Linstone (1984, 1989), Popper (1966), and Ulrich (1983), to name but a few.⁴

We call WSR a systems methodology because it is based on the ancient Oriental view of harmony, drawing upon real experiences of systems projects, manifested in a systematic form, and for the purpose of improving systems practice. WSR is to us a systems approach also because in our formulation we have recognized and tried to materialize the “new” and “higher” emergence which can only be found when parts come together to form a different whole—here we see Western and Eastern thoughts as indispensable parts of the whole of worldwide systems thinking and community.

Wuli–shili–renli is the central theme of our methodology: *wuli* denotes reg-

²We are aware that “Oriental” culture covers much more than the Chinese tradition and that the Chinese tradition during its evolution incorporated and benefited much from other Eastern traditions such as the Hindu, etc. Here I use the term “Oriental” merely to emphasize that WSR stems from the East, in contrast with those systems methodologies based on Western thought.

³For an account of the evolution and history of the WSR approach, see Zhu (1998a).

⁴I presume that most of the readers of this paper in the West have some knowledge of these achievements, therefore in this paper I pay most of my attention to exploring the relevance and significance of Chinese philosophy to systems practice. In a Chinese version of this paper, I may elaborate more on the value of Western philosophy and the systems movement in the Western context.

ularities or laws that govern the relational processes of the objective world; *shili* means manners and styles we choose to follow in order to investigate, model, and tackle natural as well as social phenomena; and *renli* concerns itself with human relations which produce unavoidable impacts on human life in general, and on systems practice in particular—especially on our ways of seeing and doing. These *lis* are differentiable yet inseparable. They constitute an irreducible whole within which each *li* cannot be fully understood and properly tackled without considering the others. The underlying philosophy of WSR is thus that *wuli*, *shili*, and *renli* will inevitably inform and condition each other, and hence will determine the fate of systems practice, whether we are aware of this or not. Ideally, in undertaking systems projects, we as systems scientists and general participants should investigate and follow these differentiated-interconnected *lis*, responding to dynamic manifestations of and among them, in each concrete situation. Our approach, WSR, tries to provide an informative and stimulating guide for assisting such research and practice [see Gu and Zhu (2000), also in this special issue, for more details of methods and process].

Surfacing and addressing *wuli*, *shili*, and *renli* together as a basic organizing schema to assist sociotechnical systems design/management appears to be a fairly recent development. However, the philosophy of WSR, i.e., the idea of a “one-ness” of interrelated principles, which humans have to investigate and to follow, has been the most deep-seated core of the Oriental culture, especially in the Confucian doctrine which is the main current of traditional Chinese thought.⁵ When I sat down to read the Confucian literature, especially the Neo-Confucianism produced in the Song–Ming dynasties (960–1644), I came to recognize and understand why systems scientists in post-Confucian societies have been insisting on the attempt, in various degrees (conscious or otherwise), to bring *wuli*, *shili*, and *renli* into the scope of their research and practice—even though a formal systems methodology for assisting such efforts might have been beyond their imaginings.

This paper is divided into three main parts. Section 2 presents an explanation of the concept of *li* and relevant teachings in the Neo-Confucian tradition. The next part, Section 3, discusses the necessity and possibility of a reconstruction of the concept of *li*. It also surfaces the current tendency in China toward scientism. Then Section 4 elaborates our theme of *wuli–shili–renli*, introducing basic concepts, discussing corresponding knowledge and methods, and warning of possible pitfalls. In the Conclusion (Section 5) I point out that in practicing WSR, we engage in encounters with the “dark side” of ancient Chinese thought and contemporary scientism.

⁵ Other traditions in ancient Oriental thought, such as Taoism and Buddhism, certainly contain valuable insights for assisting human systems management. However, since the Confucian teaching appears to me most relevant and significant to our *wuli–shili–renli* theme, in this paper I focus my attention mainly on the Confucian tradition.

2. THE CONFUCIAN CONCEPT OF *LI*

2.1. The General Meaning of *Li*

Li is one of the richest concepts in Confucian teaching, and hence in Chinese philosophy. This richness can be considered to stem from the following.

First, the Chinese language. In Chinese, a character (e.g., *li*) is often combined with one or more other character(s) to form words (e.g., *wuli*, *taoli*, etc.) which have relevant but different, or even opposing, meanings. These combined words enrich the meaning of the original key characters (Zukav, 1979, p. 32). This feature makes it possible to express more ideas and meanings in a single word in Chinese than in many other languages [for more details on this point see, e.g., Chan (1967, pp. 18–20) and Wu (1969)].

Second, since the ultimate goal of ancient Chinese philosophy was moral cultivation or self-realization rather than demonstrating the validity of one's own thesis, long elaboration or rigorous definition was considered to be unhelpful. Indeed, these were seen as limiting and undermining the value of scholarship. Rather, in the Chinese tradition, simplicity means sincerity. Therefore, Chinese classics generally took the form of brief conversations, short axioms, etc., rather than complicated arguments. For example, *The Analects of Confucius*, a book which has greatly influenced Chinese, Japanese, and Korean thought, consists only of short aphorisms and brief dialogues: *Tao Te Ching*, a book which initiated the whole Taoist philosophical tradition, is no longer than 6000 words. As they are simple, Chinese philosophical concepts and teachings are highly "suggestive" and creative (Fung, 1948, p. 14), providing significant flexibility, which in turn leaves enough room for diversified concrete experiences to be grasped simultaneously. Wu (1972, pp. 13–14) called this character of the Chinese philosophy "methodological simplicity."

Another reason for the richness of the concept of *li* may be that the whole doctrine of traditional Confucianism was based on the study of just a handful of ancient *Ru Jings* (Confucian classics),⁶ a study that allowed and required diverse explanations and elaboration of concepts, for the purposes of teaching and learning. These explanations and elaborations inevitably generated quite different meanings for the same word, due to differences in the background and interests of the commentators, as well as the influence of different historical situations. A good example in this regard is the Confucian concept of *ge*. It was reported in *Jing Yi Cong Chao* (*Collection of Excerpts in Explanation of the Classics*) that there are 72 interpretations of that concept (Chan, 1967).

⁶The following are generally considered the most important Confucianist classics: the *Wu Jing* (*The Five Classics*)—namely, *Si Jing* (*The Book of Songs*), *Shu Jing* (*The Book of History*), *Yi Jing* (*The Book of Changes*), *Lii Ji* (*The Book of Rites*), and *Chun Qiu* (*The Spring and Autumn Annals*); and the *Si Shu* (*The Four Books*)—namely, *Lun Yu* (*The Analects of Confucius*), *Meng Zi* (*The Book of Mencius*), *Da Xue* (*The Great Learning*), and *Zhong Rong* (*The Doctrine of the Mean*).

Finally, since *li* is a central concept in Confucian teaching, especially in the Song–Ming Neo-Confucianism, it has been historically used, with other relevant concepts, to construct a wide range of philosophical category pairs (e.g., *tao* and *li*, *tienli* and *renli*, general *li* and particular *lis*, *li* and *qi*, *li* and *xing*, *li* and *shii*, *li* and *yu*, etc.), which in turn were used as epistemological constructs to investigate the nature and principles of humanity as well as the natural world. Such constructions have had the effect of bringing further richness and flexibility to the concept of *li* [for those category pairs see, e.g., *The Great Dictionary of Chinese Philosophy* (Fang, 1994)].

Important and rich as it is, though, *li* seems to be a word without a definite English translation. In different contexts and for different purposes, *li* may be rendered in English as a noun, which denotes texture, marking, line, grain, form, pattern, essence, principle, law, order, rule, regulation, maxim, logic, truth, ground, argument, reason, way, what is proper, etc.; or as a verb, which denotes to manage, to run, to put in order, to tidy up, to pay attention to, to acknowledge, to distinguish, to rectify, to conduct, to serve, to engage, and so on. In *Jing Ji Chuan Gu (Terms in Classical Texts Explained)*, Ruan Yuan (1764–1849) reported that in classical sources there were more than 10 different major meanings of *li* (see also Hu Shi, 1953).

Investigating some category pairs in ancient Chinese philosophy might help us to grasp the meanings of *li*. In the pair of *li* and *qi*, *li* means principles and regulations under which matter forms and changes, while *qi* denotes the basic substance that constitutes the whole range of matter in the Universe. In *li* and *qii*,⁷ *li* means the general nature of things, while *qii* denotes differentiated and changing manifestations. In *li* and *shii*, *li* means general patterns or common ways of historical development, while *shii* denotes specific trends and particular directions at a given moment. In *li* and *yu*, *li* means should-be or ought-to-be moral norms for (or the states of thought of) sages, while *yu* denotes could be, human, unreflective, generally evil, desires or wishes.

It is not the intention of this paper to provide a comprehensive elaboration of the meaning of *li* as it has appeared in the long evolutionary history of Chinese philosophy. In the following, my introduction is limited to the extent which I consider relevant and necessary to our purpose of laying a philosophical foundation upon which to formulate a practical systems methodology for management and action research. Therefore, my introduction will inevitably be incomplete and selective.

Although the word *li* appeared in some of the ancient Confucian classics,

⁷This *qii*, which should be spelled *qi*, is another Chinese character with the same pronunciation but a different drawing and meaning compared with the *qi* in the previously mentioned pair, thus I spell it *qii* in order to signify the difference. In the rest of this paper, I adopt the same approach in similar situations.

li as a philosophical concept played no important role in Confucius's time. In ancient Confucian classics, *li* meant to manage, to put in order. It meant pattern or procedure, but not principle or law. In sharp contrast, the idea of *li* as natural or moral principle was strongly emphasized by other ancient Chinese philosophical traditions, e.g., the schools of *Tao*, *Mo*, *Fa*, *Yin Yang*, *Ming Shii*, etc., and later by Chinese Buddhism (Chan, 1967).

Rather, the ancient Confucianists placed more emphasis on *tao*. While in Taoism, *tao* was the Way of Nature (invisible, inaudible, vague, elusive); in Confucianism, *tao* was the way *of* Man, *for* Man, and *by* Man—able to be known, studied, taught, and followed. More precisely, to Confucius (551–479, B.C.), *tao* was the way of the ancient kings and sages, the way of conscientiousness and altruism, the way of wisdom, love, courage, and other virtues. “*Tao* is not far from Man,” said Confucius (*The Doctrine of the Mean*, /13). It is Man, he told the Chinese, that can make *tao* great and not *tao* that can make Man great (*The Analects of Confucius*, /15). For an individual to become a sage in society and in the Universe, the most important thing is to have *tao* (*ibid.*, /5 and /8). *Tao* in Confucianism is basically centered on humanity and morality.⁸

I present a brief introduction to the Confucian idea of *tao* here because, during the evolution of Neo-Confucianism, *tao* has come to be exchangeable with, and almost-identical to *li*, to denote patterns and principles of humanity as well as nature. To know the idea of *tao* in ancient Confucianism will help us to understand better the concept of *li* in Neo-Confucianism.⁹

It is important to note that only in Neo-Confucianism (which emerged during the 11th and 12th centuries) and afterward had *li* been granted great significance in Chinese thought. Under the impact of, and incorporating insights from, Taoism and Buddhism, the Neo-Confucianists seized upon the concept of *li*, made it their central principle, further enriched its meaning, and advanced their entire doctrine using it [this is why the main current of Neo-Confucianism is also called *Li-Xue* (The Learning of *Li*) or *Tao-Xue* (The Learning of *Tao*)].

During the hundreds of years of evolution of Neo-Confucianism, the Song–Ming Confucianist scholars borrowed and elaborated the concept of *li* generally to express their ideas of (1) regulations or patterns according to which “things” (that is, objects, events, and phenomena) actually are, shall be, or ought to be; (2) principles or ways which Man’s thinking and acting should take; and sometimes even (3) the nature of things and humans themselves. In the following, I present the three most influential streams of thinking in Neo-Confucian-

⁸This argument can be further justified by a study of another central ancient Confucian concept, *Renn*, which denotes benevolence, kind-heartedness, mercifulness, humanity.

⁹Nowadays, in the contemporary Chinese language, both in academic circles and in general life, *tao* and *li* are spoken together as *taoli* to mean principles and reason. It is no longer usual, outside the study of ancient philosophy and religion, to use these two words separately.

ism, namely, Zhang Zai's school of *qi*, Cheng-Zhu's school of *li*, and Lu-Wang's school of *xin*, which are distinctive from, but related with, each other.

2.2. Zhang Zai's *Qi* (Matter) and His Organismic View of "Oneness"

Let us begin with Zhang Zai's (1020–1077) *Qi Xue* (The Learning of *Qi*, or The School of *Qi*). Zhang Zai believed that Man and all other things are but parts of one great body, the Supreme Ultimate, constituted by one and the same primary undifferentiated material, *qi* [wandering gas, material force, or matter-energy, as Fung (1948), Chan (1957), and Needham (1956) translated, respectively]. According to this view, to become a sage, what a man should do is to remember the "oneness" of himself and all things, and act accordingly with sincerity and attentiveness. In Zhang Zai's words, a sage will serve *Qian* and *Kun* (Heaven and Earth) as we do our own parents, treat all men as we do our brothers, and extend the virtue of filial piety through service to all things.

Zhang himself did not push forward the concept of *li*. However, his harmonious "oneness" cosmology directly pointed to the necessity of, and hence gave birth to, that concept, since if the appearance and disappearance of all things are the condensation and dispersion of *qi*, then the question arises: How is it possible to explain the existence and manifestation of various categories of things? For example, why a flower is a flower and a leaf is a leaf, given that both flower and leaf are constituted by the same *qi* (Fung, 1948, p. 284).

2.3. Cheng-Zhu's *Li* and "Investigating Things/Affairs for Their Utmost *Lis*"

The above question was studied and addressed by the Cheng-Zhu School, i.e., *Li Xue* (The Learning of *Li*, or The School of *Li*). Their answer was basically that the Universe as we see it is a result not only of *qi* but also of *li*. Different categories of things exist because the condensation of *qi* takes place in different ways in accordance with different *lis*. Therefore, according to *Li Xue*, a flower is a flower because it is the condensation of *qi* taking place in accordance with the *li* of flowers, and a leaf follows the *li* of leaves. Below, I introduce first the teaching of the Cheng brothers and then that of Zhu Xi.

Both of the Cheng brothers maintained that "all things under heaven can be understood by their *lis*. As there are things, there must be *lis* of their being. Everything must have its *li*" (*Yi Shu*, 1/3b), and "There is *li* in everything, and one must investigate *lis* to the utmost" (*ibid*, 18/5b). We see that to the Cheng brothers, the concepts of *li* came to mean an unalterable principle throughout the Universe: "It is the unchangeable principle" (*ibid*, 2A/19a), unchangeable in space and time and therefore transcends particularity or events (Chan, 1957). This idea of *li* was followed by the main current of Neo-Confucianists

and was established as a central principle of the whole doctrine of Neo-Confucianism.

However, the Cheng brothers elaborated *li* along different directions. The elder brother, Cheng Hao (1032–1085), made no distinction between the above form and the within form, between general *li* and particular *lis*, saying that “concrete things are the *li*, and *li* is concrete things” (*Yi Shu*, 1/3b) and that “we say that all things are one reality, because all things have the same *li* in them” (*ibid.*, 2A/15b), while the younger brother, Cheng Yi (1033–1107), contended that *lis* for different things are different. To Cheng Yi, for a thing to exist, first there must be a certain *li* of it. For each kind of thing, there is a distinct *li* which makes it what it is. As new things appear, new *lis* are realized. Therefore, according to Cheng Yi, *li* is at the same time general and particular.¹⁰ His idea was further developed by Zhu Xi.

Zhu Xi (1130–1200) built his whole thesis on the concept of *li* as principle and reason. To Zhu Xi, for every thing, whether it be natural or artificial, there is its *li*. He stated, “When a certain affair is done, that shows there is a certain *li*” [*Zhu Zi Yu Lei (Classified Recorded Sayings of the Master Zhu)*, /101]. Therefore, for Zhu Xi, when an individual thing comes into existence, a certain *li* is inherent in it, which makes it what it is and constitutes its nature.

Regarding the relation between *qi* and *li*, Zhu Xi integrated the thinking of Zhang Zai and the Cheng brothers, saying that

in the Universe, there are *li* and *qi*. *Li* is the *tao* that pertains to “what is above shapes”, and is the source from which all things are produced. *Qi* is the material that pertains to “what is within shapes”, and is the means whereby things are produced. Hence men or things, at the moment of their production, must receive this *li* in order that they may have a nature of their own. They must receive this *qi* in order that they may have their bodily form. (*Reply to Huang Tao-fu*, in *Collected Literary Writings*, /58)

Therefore to Zhu Xi, on the one hand, “What are called *li* and *qi* are certainly two different entities” [*Zhu Zi Quan Shu (The Complete Works of Master Zhu)*, 49/5b], yet on the other hand, “In the universe there has never been any *qi* without *li* or *li* without *qi*” (*ibid.*, 49/1a). *Li* and *qi* are distinguishable, asserted Zhu Xi, yet not separable. When I read Zhu Xi, what struck me most was his argument that there is no absolute separation of *li* and *qi*, form and matter, the above form and the within form.

Zhu Xi is commonly considered as an idealist (in the philosophical sense of

¹⁰The distinction between general *li* and particular *lis* in Cheng Yi and in the whole of Neo-Confucianism is compatible with that between *tao* and *li* in ancient Chinese philosophy. In ancient times, Han Fei (280–233 BC) asserted that “everything has its unique *li*, of which the total comprehends *tao*” (*Han Fei-Tze*, /20). As I see it, general *li* in Neo-Confucianism can be seen as similar to Han Fei’s *tao*, while particular *li* in Neo-Confucianism can be considered identical to Han Fei’s *li*.

the term) and condemned as such, especially in contemporary China, because he held that between *qi* and *li* it is *li* which comes first. It is true that that was Zhu Xi's saying; for example, Zhu Xi stated, "There is *li* before there can be *qi*" (*ibid.*, 49/6a), "*Qi* is produced after there is the *li* for it" (*ibid.*, 49/6a), "Before Heaven and Earth existed, there was first of all *li*" (*ibid.*, 49/3a). But Zhu Xi also emphasized that "fundamentally *li* and *qi* cannot be spoken of as earlier or later, . . . *li* is not a separate entity. It exists in *qi*; without *qi*, *li* would have nothing to adhere to" (*ibid.*, 49/1b). "We must not say that there is *li* today and then *qi* tomorrow" (*ibid.*, 49/5b), and "Do we really know that *li* is first and *qi* later or that *qi* is first and *li* later? It is impossible to investigate this matter" (*ibid.*, 49/2b).

It seems, at first glance, that Zhu Xi was caught by the puzzle of the "earlier" and "later" and, hence, faced a dilemma of whether or not to commit to idealism.¹¹ I suggest that in order to penetrate Zhu Xi's dilemma, as we take it to be, we could investigate what Zhu Xi meant by *li* in the above quoted sayings. I also consider that for doing so the following records from Zhu Xi are informative:

(Q): How can dried and withered things also possess their own nature? . . . (A): They all possess *li* from the first moment of their existence. Therefore it is said: In the universe there is not a single thing that is without its nature. . . . For the bricks of these steps there is the *li* of bricks. . . . For the bamboo chair, there is the *li* of the bamboo chair. You may say that dried and withered things have no life or vitality, yet among them, too, there are none that do not have *li*. (*Zhu Zi Yu Lei*, /4)

(Q): Do things without feeling also possess *li*? (A): Most certainly they possess *li*. For example, a ship can go only on water, while a cart can go only on land. (*Ibid.*)

(Q): Is there *li* in dried and withered things? (A): As soon as a thing exists, the *li* is inherent in it. Even in the case of a writing brush—though it is not produced by nature but by man, who takes the long and soft hairs of the hare to make it—as soon as that brush exists, *li* is inherent in it. (*Ibid.*)

It seems evident to me that what Zhu Xi was talking about, and what was denoted by *li*, is not the ontological origin or essence of "things," or the "supreme absolute force" which produces the Universe, but natural organizing patterns and

¹¹ I must say that the debate between idealism and materialism, which is fundamental to Western philosophies, was not a central concern of, and did not have priority in, traditional Chinese thought. Of course we can pick up arguments and statements we consider relevant to the debate from ancient Chinese literature. But this cannot justify the claim that the demarcation between materialist and idealist was a "fundamental" and "prior" question in ancient Chinese philosophy. That claim, to me, is a misunderstanding and distortion introduced by both Chinese and Western writers. This distortion has seriously blocked the endeavor of learning from ancient Chinese culture. A relevant argument is given by Zhao (1995), where the author challenges the "fundamental," "prior," and "big" questions, e.g., idealism vs. materialism, which are predetermined and imposed on philosophy by "imported" Western thought. In this debate, I consider the Chinese words of *zhu* (host) and *bin* (guest) a pair of useful metaphors for grasping the "true" meaning of subject-object, idealist-materialist, in Chinese thought (see Zhu, 1998b).

principles that underlie and govern the constitution of different and rich things. Thus Zhu Xi writes, “There are *li*, even if there are no things. In that case there are only such-and-such *li*, but not such-and-such things” [*Zhu Wen-Kung Wen Ji (Collected Literary Writings of Zhu Xi)*, /46].

I have devoted so much space in the above to presenting Zhu Xi’s ideas of *li* and its relation with *qi* because I believe that, together with Zhang Zai’s harmonious view of *qi*, Zhu Xi’s teaching of *li* developed a kind of positivism which affirms the reality and principles of things and affairs, as well as empirical study, which resulted in a rationalistic intellectual doctrine, the insight of which is crucial for the WSR approach (in more general terms, the *Qi Xue* and *Li Xue* Schools brought to the Chinese scientific tradition a fundamentally new outlook, of which the influence can be found even in Japan and Korea).¹²

That doctrine of Cheng Yi and Zhu Xi (hereafter referred to jointly as Cheng-Zhu) is the doctrine of *Ge Wu Qiong Li* (the investigation of things/affairs in search of their utmost *lis*), which has supreme significance to Neo-Confucianism. Actually it was established as the only way to realize the central ideal of the Confucian tradition and to realize the ultimate goal of moral life, i.e., the ideal and goal of “sageliness within and kingliness without,” or “cultivating the self, regulating the family, governing the State, and pacifying the world.”

What does “the investigation of things and affairs in search of their utmost *li*” mean, and is it possible? To this, Cheng-Zhu had the following to say:

The meaning of the expression “the achievement of the knowledge depends on *ge wu*” is this: if we wish to carry out knowledge to the utmost, we must investigate the principles of all things we come into contact with, for the intelligent mind of man is certainly formed to know, and there is not a single thing in which its principles do not inhere. It is only because not all principles are investigated that Man’s knowledge is incomplete. For this reason, the first step in the education of the adult is to instruct the learner in regard to all things in the world, to proceed from what knowledge he has of their principles, and to investigate further until he reaches the limit. After exerting himself in this way for a long time, he will suddenly find himself possessed of a wide and far-reaching penetration. Then the qualities of all things, whether external or internal, the subtle or the coarse, will be apprehended, and the mind, in its total reality and in its relations to things, will be perfectly intelligent. This is called the investigation of things. This is called the achievement of knowledge. (*Commentary on The Great Learning*, /5)

Tang, a chief contemporary Neo-Confucianist, gave a good explanation of the knowing of particular *lis*, thus:

¹² As Needham (1956, p. 412) summarized, “The Neo-Confucians arrived at what was essentially an organic view of the universe. Composed of matter-energy (*qi*) and ordered by the universal principle of organisation (*li*), it was a universe which, though neither created nor governed by any personal deity, was entirely real, and possessed the property of manifesting the highest human values (love, righteousness, sacrifice, etc.) when beings of an integrative level sufficiently high to allow for their appearance, had come into existence.”

Although *li* may be a universal, a particular, or simply a property of an individual thing, when Cheng I-chuan [i.e., Cheng Yi] talked about *li* he was more concerned with the *li* of particular and individual things in ordinary life than with the universal principle of the universe as a whole. Here Cheng I-chuan paid more serious attention to the fact that our ordinary life takes place in concrete situations composed of particular things. So Man has to search for and grasp the forms, patterns, principles, and laws, as the *lis* of his own thinking and acting which are necessary for his having true knowledge of these things, and for his right and good action in terms of the proper response to these things. (Tang, 1972, p. 60)

It is only through the investigation of concrete, specific, and particular things and affairs that Man can know what the proper response should be, Cheng-Zhu maintained. Searching for not only the general *li* of the metaphysical principle of the Universe but also particular *lis* manifested in particular things and affairs, so that we can make right and good responses in our everyday concrete situations, is therefore the *li* of human life. In searching for the utmost *lis* of things and affairs, Man simultaneously obtains the *li* of his life. Here we find that, during the process of “investigation of things and affairs in searching for their utmost *lis*,” the *lis* of things and affairs and the *li* of human life come to be identical, integrated, and to be but one. We can say, arguably, that the idea of such integration constitutes the positive epistemological dimension of *Tian Ren He Yi* (The unity of Heaven, i.e., the Universe and Man)—the Confucian view of harmony.

For the Neo-Confucianists, especially for the Cheng-Zhu school, as *li* means the way or road that Man’s thinking and acting can take, then it is necessary and possible to distinguish between real *li* and impasse. The real *li* for Man’s thinking and acting is one that responds correctly to the *li* of things and affairs, and is therefore the way man *should* go, and also the only way Man *can* really pass along and through. Like *tao* in ancient Confucianism, *li* here is identical to the way of human life, reason, and norms of behavior, reflecting a belief that Man has an innate capacity to think and act with his reason as an embodiment of *li*. This *li*, which any person who wishes to cultivate and actualise himself should search for and follow, is the *li* of his life *and* of the world. Thus we have the Cheng brothers” saying that “there is only one *li* in the world. You may extend it over the Four Seas, and it is everywhere true” (*Yi Shu*, 2A/19a). So much so that Cheng-Zhu stated that “*xing* (human nature) is *li*,” through which, and only through which, *li* was granted an ontological characteristic: *Li* is not only the most general schema of the known object, but also the inherent nature of human beings as knowing subjects.¹³

The Neo-Confucianists believed that searching for particular *lis* of specific

¹³In ancient Confucianism, *li* remained mainly moral-philosophical. Only since Neo-Confucianism, especially the Cheng-Zhu school, has *li* also been granted an ontological character. For more discussion on this point see, for example, Cai (1995) and Lu and Cheng (1993).

things and affairs in concrete situations is not only ideally necessary, but also practically possible. As a contemporary Neo-Confucianist stated,

One might even say that due to their great, even infinite number, the investigation of all things is an impossible requirement for sagehood. ... [I]f Man's investigation of things in his search for their *lis* was merely a quest for theoretical knowledge of objective things themselves, ... then it surely would be an endless enterprise. But if it is for the purpose of knowing the *lis* of our right and good responses to things in concrete life-situations, it is not a hopeless matter. The things a man encounters in concrete situations are limited, and the *lis* of right and good response that he should know may be even less. ... In a particular situation, ... the investigation of things with a view to finding their *lis* is what Man can and should do. (Tang, 1972, p. 70).

It is in this idea of searching for particular *lis* for right and good responses toward concrete situations in our everyday life that I find one of the most valuable insights of Neo-Confucian teaching. That is, due to the limited capacity of human beings, we as systems scientists or whoever, may not be able to reach the ultimate "truth" of the Universe, the world, the metaphysical, no matter how sophisticated and how advanced the methods and technologies we employ. However, when we turn our attention to our real, here-and-now situations; or in Western jargon, when we need to intervene in sociohistorically shaped, local and temporal situations; or when we are to undertake social problem-solving for the purpose of improving such situations; it is desirable, necessary, and possible, as the Neo-Confucianists tried to convince us, to search for and arrive at differentiable *lis* which we can and should follow to make appropriate here-and-now responses.

Given that knowing *lis* of things and affairs is not only necessary but also possible, one might still ask, Why do many people know *li* yet fail to follow it? How is it possible for our response to follow *li*? These questions were addressed substantially by Zhu Xi. To Zhu Xi, it is not enough for Man to have a will to investigate things in searching for their *lis*, he should also have a will to realize the *lis* he knows. Once he knows the *lis* of things, he should take them as a standard of his response. Actually, Zhu Xi argued that we must *first* have a will which actually likes good and dislikes evil. Here Zhu Xi leads us back to one of the teachings in *The Great Learning*, which suggested the need to make "the (moral) will real." Therefore Zhu Xi, once again, reminds us of the importance and priority of the moral dimension in searching for *lis* (*ibid.*, p. 75).

Furthermore, Zhu Xi recognized and discussed substantially the partiality of mind, the causes of this partiality, and the way we can tackle it.¹⁴ Our natural feelings or emotions which attach to things and affairs, Zhu Xi maintained, have

¹⁴This reminds us of Kant's (1929, 1951) thesis of incomplete human knowledge as well as Ulrich's (1983) thesis of partiality, selectivity and "boundaries" in social systems design. Zhu Xi's argument remained basically psychological, failing to formulate an operational methodology or guideline. But let us not forget that Zhu Xi lived early in the 12th century, when priorities were different.

particular directions. These directions, in turn, produce partiality and bias in our will. Therefore it is necessary to put our mind in its right place and point our will in the right direction through a process of reflecting and mastering our feelings and emotions, so that the will, springing from the mind, will be directed only to the good and right. Doing so, we are able to conform to the *lis* we know, and thus will be able to make good and right responses to outside things and affairs, as well as to realize the *li* as the inner nature of Mankind. In his elaboration, Zhu Xi connected the process of overcoming partiality of mind toward moral will, again, with the teaching in *The Great Learning*; this time, the teaching of “rectification of mind” (Tang, 1972, p. 76).

2.4. Lu-Wang’s *Xin* (Mind) and Freedom of Ethics

Later, *li* was further identified with the human mind which embraces the self, others, and the Universe, and therefore was eventually granted the status of Subject. This was done by Lu Jiuyuan (1139–1192) and Wang Yangming (1472–1529) (hereafter Lu-Wang), especially the latter, who contributed most to the development of *Xin Xue* (The Learning of Mind, or the School of Mind), another major school of Song–Ming Neo-Confucianism.

Lu-Wang was uncomfortable with the dualist nature of *li*, as they took it to be, in the Cheng-Zhu school. The Cheng-Zhu school was dualist, according to Lu-Wang, because their idea of the separation between the will to know and the will to follow *li* would inevitably leave room for separation between the search for the *lis* of things/affairs and the inquiry into the *li* of human life. There are three major theories in Lu-Wang’s school: namely, the monistic philosophy of *Xin* (Mind), the doctrine of the unity of knowledge and action, and the theory of realizing conscientious wisdom.

Rejecting Zhu Xi’s attempt to seek knowledge in and from objective phenomena in the external world, Lu-Wang proposed a system of idealistic monism: “*Xin is Li*” (My mind is the Way) [*Lu Xiang Shan Quan Ji (Collective Works of Lu Hsiang-Shan)*, /12], “The Universe is my mind; my mind is the Universe” (*ibid.*, /33). This is a system in which the mind, its intention, its knowledge, and its objects together form a manifold scheme of inseparability. It is another “oneness,” a “oneness” which is of course fundamentally different from that of Zhang Zai, as well as Zhu Xi. In this conception, the Universe is now a spiritual whole, in which there is only one world, the world that we ourselves experience mentally. Thus there is no place for ‘that other world’ of things/affairs, which Zhu Xi so much emphasized (Fung, 1948, p. 309).

To the Lu-Wang school, *Li* as reason is the function of Mind. There is no *li* of acting or response outside the mind and belonging to so-called ‘objective things/affairs’ alone. Lu-Wang went further to claim that the *li* in our mind is at the same time the mind’s *being* conscious and the *li* of which the mind is

conscious. That is, Mind is the unitive core of both the substance and function of human nature.

Wang Yangming continued to state that “there can be no objects outside of mind, no events beyond the reach of mind, no reason independent of mind, no righteousness apart from mind, and no goodness in the absence of mind” (*Chief Works of Wang Yangming*, 4/14). Therefore we see that in *Xin Xue*, the investigation of things came to mean investigating the objects in the mind, investigating the objects in intention, and investigating the objects in cognition only (*ibid.*, 4/26).

It appears clear that the whole doctrine of Lu-Wang’s *Xin Xue* is idealist. However, we can derive a valuable insight from it; that is, the insight of its idealist “oneness.” It is through “My mind is the Universe,” according to Lu-Wang, that the separation and even the distinction between the mind and the world, the subject and the object, the self and others, no longer exists or has meaning. Through our mind, we reach the ultimate harmony of Man and the Universe: *Tian Ren He Yi* (The unity of Heaven and Man), which is a realm where the self is no longer a small self, but a big self which embraces others: other men, other creatures and all other things. The self becomes the Universe and the Universe becomes the self. Perceived as such, people should, naturally, extend their boundless sympathy to others. What a person does to him or herself is done at the same time to other people, as well as to all other creatures and things. For the good of others and for the Universe, what such a self needs to do is not to overcome or to constrain the self, as the conventional interpretation of the Confucian classics would have us to do, but to pursue infinite enlargement and endless edification through the self’s creative action. In Wang Yangming’s words,

To manifest the illustrious virtue is to establish the nature of the unity of Heaven, Earth, and all things; to love people is to exercise the function of that unity. Therefore the manifestation of the illustrious virtue consists in loving people, and to love people is to manifest the illustrious virtue. If I love my own father, the fathers of some other men, and the fathers of all men, my love will be truly extended with my love of these fathers. . . . Beginning with all these human relationships, and reaching to mountains, rivers, spirits and gods, birds and beasts, grasses and trees, all should be loved in order to extend our love. In this way there is nothing that is not manifested in our illustrious virtue; and then we are really one with Heaven, Earth and all things. [*Da Xue Wen (Questions on the Great Learning)*, in *Complete Works of Wang Shou-Ren*, /26]

It is through this thesis of “the unity of Mind and Universe” that the *Xin Xue* school of Neo-Confucianism granted *li* with the status of both the object and the subject, encouraging the pursuit of creativity through Man’s knowing of, and acting on, the “oneness” of the self and others.

In Zhu Xi’s thought, “making the (moral) will real” is a distinguishable step, to say the least, from that of knowing *li*. But for Lu-Wang, the two are actually one.

Criticizing Zhu Xi, Lu-Wang maintained that the proper understanding of *li* should be that the feeling of liking good comes no later than knowing good, and the feeling of disliking evil comes no later than knowing evil. Wang Yangming argued that knowledge is the chief plan for action and action is the actual work done in knowledge; knowing is the beginning of acting and acting is the accomplishment of knowing (*ibid.*, 2/I: 6). Therefore, to Lu-Wang, knowledge and action constitute one inseparable unity; that is, *Zhi Xing He Yi* (knowledge and action, knowing and doing, are but one). We see that while Zhu Xi maintained that knowledge and action require each other, Lu-Wang advocated that knowing *li* and following *li* are actually the “same thing.” In history, Lu-Wang’s theory of the unity of *Zhi* and *Xing* (knowing and doing), together with the theory of “Mind is *Li*,” educated and encouraged the Chinese to fight for good against evil, and encouraged the Japanese to fight against evil rulers. Its value, it seems to me, is still apparent today. As Fang (1973, p. 77), another contemporary Neo-Confucianist, states,

Such a theory has an especial significance for ethics. Nowadays, scholars, owing to the acceptance of the duality of knowledge and action, will not try to nip in the bud that idea of ill-intention just because of its being looked upon as a mere idea which, according to Wang Yang-ming, is already a scheme of action and has got to be uprooted from the very beginning before issuing in explicit evils.

The concept of *liang-zhi* [conscientious wisdom, or intuitive knowledge, as translated by Fung (1948) and Tang (1972), respectively] and the doctrine of *Zhi Liang-Zhi* (realizing conscientious wisdom) are independent achievements of Wang Yangming, which can be expressed as follows:

In the teaching of Wang Yangming, every man has *liang zhi* because he has a consciousness of some *lis* which should be followed, and thus has some knowledge of what should be, . . . We need not know all of what ought to be, or all *lis*. They will be revealed gradually in the process of our lives. We are ordinary men not because we have no knowledge of what ought to be, but because we do not act fully upon the knowledge we have. We are not responsible for the “should be”s we do not know. Our responsibility consists in responding to any “should be” that is immediately expressed, revealed, and known. This is quite enough for our immediate moral practice. . . . this consciousness of *li*, or *liang-zhi*, is originally both knowing and feeling. The fact is that when *liang-zhi* knows the right or good, a feeling of liking it arises simultaneously, as does, when *liang-zhi* knows the wrong or evil, a feeling of disliking it. (Tang, 1972, p. 80)

To Wang Yangming, every person has *liang-zhi*, therefore every person has the potential to become a sage. One can become a sage if one but follows the dictates of one’s *liang-zhi* and acts accordingly. Wang’s thesis of *liang-zhi* has therefore greatly developed Confucius’s teaching that “to be benevolent is up to the self” (*The Analects of Confucius*), which emphasizes the inherent freedom and consciousness of human beings who may study and practice ethics.

In this section I have presented a brief introduction to the Neo-Confucian

teaching, with an emphasis on its relevance to our WSR theme. The evolutionary thread of Neo-Confucianism, to me, is clear. It started as a philosophy of searching for *lis* of the harmonious oneness, then modified itself into a philosophy of human nature which is identical with the *li* of that oneness, and eventually transfigured itself into a philosophy of *xin* (mind) which encourages humans to realize *liang-zhi* for the self and others. In spite of great variations and critical debates among the three major schools, there is a point of convergence among them all. That is, they are all committed to searching for and following *lis*, whether as natural patterns, as Man's reason, as his intuitive knowledge, or even as the interplay and integration of all of them. And this ideal has been established as the Way to become an inner sage and outer king, i.e., to actualize the self and to benefit society.

3. REFINING AND RECONSTRUCTING *LI*

3.1. Valuable Insights

The concept of *li* and relevant doctrines in Chinese culture contain, in my view, many stimulating and inspiring insights for systems thinking, among which I consider the following, although subject to disagreements and controversies, particularly relevant and significant for the formulation and practice of WSR.

(1) *The harmonious view of "oneness."* We see the idea of "oneness" which embraces the self and all things in the world of Zhang Zai's *Qi*, Cheng-Zhu's *Li*, and Lu-Wang's *Xin*. The Neo-Confucian conception of the Universe is that of a single organismic unity. All things and affairs exist in dynamic relations, and all relations follow a dynamic mutual-causal principle according to which things/affairs are organized. In this respect, Needham (1956, p. 458) concluded that Neo-Confucianism "was fundamentally a philosophy of organism." He was also happy to translate *li* as "Organization" or "Principle of Organization" (*ibid.*, p. 457).

(2) *The desire for freedom of the subject and autonomy of ethics.* Particularly in Lu-Wang's school, it is claimed that "all affairs within the Universe come within the scope of my duty; the scope of my duty includes all affairs within the Universe" [*Lu Xiang Shan Quan Ji (Collected Works of Lu Xiang Shan)*, p. 33], from where we see a highly developed self-consciousness of ethical responsibility.¹⁵ Furthermore, this consciousness of responsibility does not lead to overcoming or constraining the self, but encourages the self to pursue infinite enlargement and creativity. Every person has his or her original good will and conscientious wisdom; therefore every person has the potential to become a sage, as long as s/he elects to do so. "The streets are full of sages," Wang Yangming and his followers said. That is, every person has freedom and autonomy, and is full of great potential and creativity, in the practice of ethics and morality.

¹⁵I suggest that this ideal can be traced back to the ancient time. As Mencius (372?–289? BC) said, "All things (literally "ten thousand things") are complete within me" (Mencius, VII/I).

(3) *The belief in the unity of knowledge and action.* For the Cheng-Zhu School, the purpose of investigating things in search of their utmost *lis* is to follow those *lis*, while for the Lu-Wang School, knowing conscientious wisdom is realizing it. According to Zhu Xi, knowing *lis* and following *lis* require each other, while for Wang Yangming, knowing *lis* and practicing *lis* are actually one. For Neo-Confucianism as a whole, to sage (i.e., to actualize) the self is at the same time to king (i.e., to benefit) society.

(4) *The pursuit of methods.* Every major school formulated its own methods. Therefore we have Cheng-Zhu's "investigation of things and affairs in search of their utmost *lis*," "making the will real," "rectification of mind," and Lu-Wang's "realise conscientious wisdom" and "rectification of affairs" (Fung, 1948, pp. 313–315). With regard to searching for different ways to tackle different *lis*, it appears to me that the following message from Cheng Yi has great significance:

There are many ways of investigating *lis* to the utmost. One way is to read about and discuss truth and principles. Another way is to talk about the people and events of the past as well as the present, and to distinguish which is right and which is wrong. Still another way is to handle [human] affairs and settle them in the proper way. (*Yi Shu*, 18/5b)

When we put these insights together, an underlying logic can be discerned and becomes clear to us. First, given the "oneness" constituted by the unity of being, the unity of existence, the unity of life, and the unity of value (Fang, 1973, p. 73), we should search for both general and particular *lis*. Next, because to know those *lis* is to follow them, we should include all those *lis* in our plan and action for making appropriate responses. Then, since there are particular *lis* for particular concerns in particular situations, we need particular ways for knowing and acting. Finally, it is our responsibility, and we have full potential, to do all the above through our conscientious wisdom and creativity. It is this logic underlying Neo-Confucian teaching as a whole that Gu and I employ as the philosophical ground of our WSR approach.

But before I move to push forward the *wuli–shili–renli* theme, I discuss briefly the need for a reconstruction of the concept of *li* and hence the doctrine of Neo-Confucianism, so that WSR may not be misunderstood and misused.¹⁶

¹⁶ "Reconstruction" in its relation to Confucianism in contemporary Chinese societies is a "big" word. It is used to denote the century-long efforts of contemporary Confucianist scholars in China as well as overseas to restore the spirit and values of Confucianism and traditional Chinese culture in the face of challenges from Western thinking and the breakdown of traditional value systems. For discussions and evaluations of this reconstruction, see, for example, Xiao (1994), Cheng (1994), Xu (1994), and Zhu (1994) (all in Zhang, 1994), Fang (1993), Li (1993), Zhou (1994), and Xie (1994). In this paper, my discussion of "reconstruction" is a relatively "small" one, concentrating strictly on what I consider to be directly relevant to WSR.

3.2. The Need for Reconstruction

To move from what Confucianism *was* toward what it *ought to be*, in my view, we need to reconstruct it, especially in the following aspects.

(1) *The preoccupation with morality and ethics.* First, we need to reconstruct the *li* embedded in the Confucian view of harmony. In spite of speaking of individual *lis* and different categories of unities, the Confucian teaching fundamentally concentrated on human and social affairs, with an imbalanced interest in morality and ethics. The Neo-Confucianist harmony is essentially a cosmic-ethical or value-centric unity. It is a moral philosophy rather than a cognitive one, or more precisely, it is a philosophical anthropology (Fang, 1964, p. 103). The Confucian *Tian* (Heaven) in “the Unity of Heaven and Man” has hardly been granted the meaning of Nature (Zhang, 1995, p. 204). Rather, it was generally related to the Supreme Ultimate which determines Man’s ethics and fate (Li, 1995; Luo, 1994). Needham (1956, p. 32) criticized Neo-Confucianism, claiming that it “simply turned away its face . . . from Nature and the investigation of Nature to concentrate a millennial interest on human society and human alone.”¹⁷ Although it may be open to question whether Needham’s criticism is justified, it nevertheless seems quite safe to argue that Confucianism is centered on moral values rather than cognitive knowledge.

Accordingly, the essence in the Confucian doctrine of harmony is basically a pursuit of “the natural way” of human spiritual and moral life, rather than “the way of nature.” The primary idea underlying the Confucian concept of *li* embedded in its view of harmony is how to become a sage through cultivating the self in terms of morality and ethics. Even the ideal of “kingliness without” was, for the Confucianists, the by-product of, and the avenue toward, the ultimate goal of “sageliness within.” Thus we have Confucius’s saying that “to know how to be sincere is the way of Man” (*The Doctrine of Mean*, /13). Needless to say, “action” in Lu-Wang’s “Unity of Heaven and Man” was not really to be interpreted in terms of work and labour; rather, it was essentially concerned with the “total moral performance of the individual” (Wu, 1972, p. 10). For Confucianists, knowledge was acquired not for its own sake, but for the ultimate goal of self-sageliness. I think Needham is right in his statement that in Zhu Xi, the investigation of things was centered on the *li* of Man, with Nature considered secondary (Needham, 1956, p. 150). With its primary interest in the knowledge of ethics, Neo-Confucianism neglected and downplayed the investigation of objective things and events, and hence those corresponding *lis*. Indeed, if “*Tao* is not far away from Man,” as Confucius told us (*The Doctrine of the Mean*, /13), then why bother to investigate things in Nature

¹⁷ Another relevant message from Needham (1956, p. 579) says that “in Europe natural law may be said to have helped the growth of natural science because of its Universality. But in China, since natural law was never thought of as law, and took a very social name *li*, it was hard to think of any law as applicable outside human society.”

far away from Man?¹⁸ In my view it was the preoccupation with moral values and human affairs that prevented those brilliant scholars from paying due attention to objective phenomena, to investigate objective things and events, to search for their *lis*.

According to experiences in systems projects in the East, and as demonstrated by Habermas (1972) in the West, however, *lis* of the dynamic process of objective phenomena can be distinguished from, and are by no means less important than, moral and ethical values, for improving human situations, and for systems research and practice. I therefore suggest that the imbalance of content in the Confucian view of the undifferentiated “oneness” and the corresponding concept of *li* should be reconstructed. Accepting the importance of moral and ethical value should not lead us to ignore or to downplay the investigation of objective things and affairs.

(2) *The “closed” epistemology.* The imbalance of concerns between the self’s moral performance and external matters of fact in the Confucian view of an undifferentiated “oneness” was derived, it seems to me, from the conception of the so-called “inseparability of subject and object.” While not rejecting the positive value of this conception, especially its value for addressing current issues such as the worldwide ecological crisis, we should be aware of its inherent limitations and the negative impact it has had on ancient Chinese thought.¹⁹

“Real” epistemology can begin only when the object and the subject are conceptually distinguished. Primitive people lived within a muddy mix of object and subject, or in Popper’s (1966, Chap. 5) terms, a “charmed circle.” They did not see the differences between the two. Therefore they could not see the nature of either. Man was god(s), and gods were nature. It is the conceptual distinction between object and subject that brought human beings the knowledge of both Man and nature, self and other, value and fact, natural environment and social environment, natural laws and normative principles, primary facts and human decisions, etc. Without knowing these differences, we are bound to be imprisoned within a “value curtain” of unreflected subjectivity which has killed, and

¹⁸ Another Confucianist saying reads, “The way of Heaven is far away, while the way of Man is near” (cf. *Zi Can*).

¹⁹ Many of us seem to enjoy traditional Chinese thought, e.g., “In Chinese philosophy (not Chinese philosophy as prepared for Western readers by some scholars) we do not see the customary dichotomies such as form and matter, mind and body, idealism and materialism, *a priori* and *a posteriori*, analytic and synthetic, the empirical and the rationalistic, subjectivity and objectivity, and so on” (Wu, 1972, p. 6). “Since there is no original demarcation between the objective and the subjective, the subjective in man naturally corresponds to the objective in nature. This may be regarded as a metaphysical article of faith, but it has the virtue of ruling out epistemological puzzles about knowledge of the external world and other minds. Hence, there are no doctrines of solipsism and scepticism in Chinese philosophy” (Cheng, 1972, pp. 159–160). While I realize that there are some advantages to having “no dichotomies,” in my discussion here I would like to point to the need for awareness of the dark side of this viewpoint.

will kill, any possible knowledge about both the subject and the object (Ying, 1988, pp. 432, 467). To me, the object cannot be reduced into the subject, just as the subject cannot be reduced into the object. They require and condition each other. Both, related yet distinguishable conceptually, are indispensable preconditions of our cognitive capability.

Unfortunately, this is not the case in Neo-Confucianism. Especially in Lu-Wang, the distinction between the subject and the object is dissolved in the endless enlargement of the subject. Consequently, the other is reduced into the self, the Universe is reduced into Man, and eventually all things are reduced into Mind. Without an independent object conceptually distinguished from the subject, without necessary opposition and confrontation, the Confucian spirit lost the possibility and basis of critique, and therefore established itself as a "closed" system, within which the moral dimension was sophisticated while the political and scientific dimensions were handicapped. As a result, although exclusively centered on human life and social affairs, Confucianism could not provide concrete and practical proposals for political and scientific progress, and therefore failed to respond to the challenge of the object, e.g., the scientific power of the West. Even in the field of its central concern, i.e., human life and social affairs, it stopped at endless discussions with ethical language only. It might be able to articulate the subjective *ought to*, but was incompetent to tell the objective *what is*, because when it claimed that "my mind is the Universe," the objective world was dissolved into the subjective mind, and the cognitive system was killed by that closed value system.

To penetrate the "value curtain," to encourage the pursuit of necessary "positive knowledge," we need an adequate epistemology, which is possible and available only when the conceptual distinction between the subject and the object, between human norms and primary facts, is recognized and established. Therefore, to distinguish the subject and the object, value and fact, and corresponding differentiated *lis*, should be the second aspect in our proposal for reconstruction.²⁰

(3) *The conservative ethical idealism.* Another aspect to be reconstructed, I suggest, is the idealist trend in the Confucian doctrine of ethical autonomy. "Sorting out my mind, conforming to my value, so that everything is up to me, what else do I lack then?" said Lu Jiuyuan [*Lu Jiuyun Ji (Works of Lu Jiuyun)*, 315]. As for Wang Yangming, *li* is not in things but in mind alone. Certainly,

²⁰"The unity of Heaven and Man" in the traditional Confucianist form, without proper reconstruction, should not be taken as the panacea for our modern ecological illness. We cannot return to synthesis without analysis. At the risk of incurring unpopularity, I would like to quote Popper (1966, p. 177) thus: "We cannot return to a state of implicit submission to tribal magic. For those who have eaten from the tree of knowledge, paradise is lost. . . . *There is no return to a harmonious state of nature. If we turn back, then we must go the whole way—we must return to the beasts.*"

such ethical idealism has inspired and encouraged the pursuit of the autonomy of ethics. But how far is such ethical idealism from ethical relativism? How can we know that the *li* obtained from one's self-cultivation is necessarily good and not evil? How is it possible to reach a "oneness" among individuals toward the right values?

Neo-Confucianists either left these questions unaddressed or turned to their hypothesis of "the nature of Man is originally good" (Mencius, VIB/2). But a hypothesis is a hypothesis, no more and no less. Its fate is determined by its actual pragmatic functioning. The fact is that all "rebellions" in history, without exception, came under the name of *Ti Tian Xing Tao* (Acting upon the Way of Heaven). Every time, it fell to the victorious and the powerful to interpret the *Tao*, or *Tianli* (the Way of Heaven). More often than not, *Tianli* in Confucianism was identical to the ideology of the feudal autocracy, was defined in terms of the existing political structure and relations, and was used to demonstrate the isomorphism between Heaven and the status quo. The *Tian* (Heaven) in Confucianism was seldom related to Nature but to the Supreme Force who granted legitimacy and "the Way" to its incarnation on the earth, *Tian-zi* (Son of the God, i.e., the Emperor). This is an unfortunate dark side of the Confucianist tradition, from Confucius's (551–479 BC) "*Wei Tian-ming*" (Be dread of fate), to Dong Zhongshu's (179–104 BC) "*Tian-zi* gets the way from Heaven, and the world gets the way from *Tian-zi*" (*Han Shu*, /*Dong Zhongshu*), and then to Cheng-Zhu's "preserving the Way of Heaven and killing human (evil) desires" (*Yi Shu*, /24; *Zhu Zi Yu Lei*, /13).

As a result, ethical idealism embedded in the Neo-Confucian doctrine did not bring autonomy of ethics or lead to freedom of value selection. Rather, it was always established as a powerful ideology to reinforce the existing sociopolitical structure, which eventually stripped the autonomy of the subject, and dissolved the legitimacy of *Tian-li* itself. "*Tian-li* is killing," "*Li Xue* is killing," are the cries of grief against the authoritarianism and conservatism of Neo-Confucianism [see, e.g., the writings of Lu Xun, a modern Chinese writer, and Dai Zen (1723–1777), a critical Confucianist in the 18th century]. The freedom of ethical selection was eventually substituted by commitment to the authorised value system. Thus, ethical idealism in the form of Neo-Confucianism was no less depressing than ethical autocracy.

In order to address and to deal adequately with human relations, which are inevitably involved in systems projects, both ethical relativism and autocracy should be rejected. A way out of such ethical relativism and autocracy, it seems to me, is to move our ethical thinking from the paradigm of isolated subjectivity toward that of communicative intersubjectivity, which in the West is substantially addressed by Husserl (1970), Heidegger (1977), and, especially, Habermas (1981a, b).

(4) *The handicap of methodology.* Finally, but importantly, we must recon-

struct the Method in Neo-Confucianism. Compatible with their non-analytical “oneness” ontology, and with their undeveloped epistemology, the Neo-Confucianists’ method was seriously handicapped. In the main current of Confucianism during its whole history, “investigating things” came to mean understanding right and wrong from the ancient classics. In the record of Zhu Xi’s sayings, we see that he made certain observations on natural as well as social phenomena, but most of his time was devoted to the study of, and comment on, the classics (Fung, 1948, p. 319; de Bary, *et al.*, 1960, pp. 479–480). Those later scholars, who were critical of Song-Ming *Li Xue*, also devoted most of their energy to working with books and documents, when their contemporaries in Europe were working with natural phenomena (Chan, 1957, p. 331). As for Lu and Wang, both of them claimed that they obtained their doctrine through sudden Enlightenment. All Cheng-Zhu’s “investigation of things,” “making the will real,” “rectification of the will,” and Lu-Wang’s “realising conscientious wisdom” were exclusively colored by moral intuitionism,²¹ and therefore “failed to reach any precise definition of the experimental method, or any systematisation of its observations of Nature” (Needham, 1956, pp. 161–162). Hu Shi (1953) argued that the Neo-Confucians had the scientific spirit but not the scientific method.²² Even contemporary Neo-Confucianists criticize the methodological (as well as the epistemological) handicap in traditional Confucianism (see, e.g., Fung, 1984, p. 297; Liu, 1969, 1970, 1972). It appears safe to say that the Neo-Confucianists’ method was a synthesis without analysis, which remained primitive, could not practically provide appropriate methods for searching for different kinds of *li*s underlying various kinds of things, events, and relations, and can therefore be called “prescientific” (Yang, 1995, p. 30).

Furthermore, each Neo-Confucianist tended to commit merely to his own method. As Fung (1948) describes, according to Zhu Xi, one must in principle

²¹ Wu (1972, p. 14) argued, “It has been a general understanding that the major methodology in Chinese philosophy is *intuition*. This term is a very unfortunate one, for it has so many meanings. If it has to be used, it must not be taken to denote the kind of intuition which is primitive, impulsive, instinctive, unreflective, and anti-scientific. Nor does it denote an occult vision which is possessed only by mystics. It is the kind of intuition which is the natural synthesis of the outcomes of moral maturation, intensive observation, direct experience, and persistent intellectual effort. This kind of methodology defies any ready-made categories in Western philosophy. If I have to give it a name, I would call it “a natural synthetic intuitive grasp.” It is not antagonistic to logic or analysis.” More discussions on the intuition method in Eastern thought are given by Feigl (1958), Raju (1958), and Chang (1960). Capra (1992) also discusses the value of intuition in human knowledge inquiry. Here, my emphasis is to criticize the imbalance and handicap in the Neo-Confucian method(s). Therefore, to me, Fung Yu-Lan’s following saying seems more convincing. It reads, “... Before the negative method is used, the philosopher or student of philosophy must pass through the positive method, and before the simplicity of philosophy is reached, he must pass through its complexity. ... One must speak very much before one keeps silent” (Fung, 1948, p. 342).

²² I agree with Hu Shi on this point, although I think his definition of scientific method should be open to discussion.

start with the investigation of things, although he himself did not strictly carry this out, while the followers of Wang Yangming, on the other hand, came to believe that their intuitive method alone could itself tell us everything, including the “know-how” (*ibid.*, pp. 319–320). This kind of “imperialist” attitude toward “alien” methods has been criticized recently in the West by, for example, Flood and Jackson (1991a, b). The basic message of Flood and Jackson tells us that, on the one hand, there can be no “super method,” while, on the other, we cannot return to the way of trial and error. What we need in undertaking systems projects is therefore to select and/or to design appropriate methods through open and critical discourse among participants so as to tackle heterogeneous situations in a theoretically informed way. I consider this suggestion a feasible direction we can take in order to overcome the methodological handicap in the Confucian tradition.

3.3. The Contemporary Tendency Toward Scientism

In the above I raised several issues which I consider should be addressed in a necessary reconstruction of Confucianism. In doing so, I undertook a critique of the Confucian tradition, especially its imbalanced concept of *li* and its undifferentiated view of harmony. I criticized its exclusive concern with morality, humanity, and social affairs, which was always full of feudal ideology; I criticized its ignorance of objective things and natural principles, which led to a failure to promote positive knowledge in both the natural and the social sciences; I criticized its exclusive reliance on intuition and its imperialist attitude toward methods, which blocked different ways for studying different *lis*; I also criticized its ethical idealism and relativism, which unfortunately led to autocracy and conservatism.

If I were to stop here, some may expect that in the formulation of WSR we will be dedicated mainly to promoting and reinforcing objective cognition, fact-based judgment, empirical method, positive knowledge, instrumental techniques, and the like, or that we will commit to moving the focus of systems scientists away from ethics, morality, humanity, and social affairs. But I must say that we will not. We are only at the halfway point of our discussion. In the above, my critique was leveled at the limitations and the dark side of Confucianism as it *was*. We are only just beginning to consider what an ideal facilitating Confucianism in our present time *ought to be* if we are to restore the value of Chinese culture, and to employ it as the philosophy of a facilitating systems approach for management and action research. For this, we must enter the second half of our journey. We have to investigate and analyze what is the other, perhaps the major, pitfall in our current systems research and practice.

In my view, most of us in the systems community in the East, most surely in present China, are confronted with a realistic and serious tendency, which is

in opposition to that of the Confucian tradition I have just criticized. What I am to surface here is a new kind of undifferentiation, imbalance, preoccupation, and reduction—that of scientism.²³

By the term scientism, I mean the belief that all aspects of the Universe and human life are knowable and manageable through the methods of experimental science. Indeed, scientism rules all “unscientific” knowledge automatically invalid.

In the East, especially in China, scientism emerged around the turn of this century, and has gradually come to prevail since then. Due to the achievements of the modern sciences in the material and political (especially international political) spheres, contemporary scientists have generally come to believe that *the* scientific method is the only way to study material as well as spiritual phenomena, and by “scientific method” they mean exclusively positive-empirical methods which were generalized from laboratory experience. For example, Hu Shi, a modern Chinese scholar, claims that “scientific method” is not only adequate for investigating all phenomena, i.e., things and events, but also appropriate for analyzing value issues such as human worldviews. Also, Ding Wenjiang, a modern Chinese scientist, argues that “the power of science is not in its content, but in its method” (Gong, 1994). Such scientism has had a serious influence on the general public as well as academicians.

There are historical reasons for the prevalence of scientism in China. Partly, it is due to the general failure of Chinese intellectuals to make the transition from the past to the present a smooth one. In the continuous turmoil of political, economic, and social upheaval, especially when the nation was confronted with challenges from the *Xi-yang* (the West) and the *Dong-yang* (the East, i.e., the Japanese),²⁴ there was little time for cool and rational analysis, evaluation, and comparison between materialist power and spiritual values, science and ethics, preserving our own traditions while learning from others, etc. Initially, in this transition, it was claimed that there were two equally important figures to be invited from the West: Mr. S (science) and Mr. D (democracy). However, due partly to the more immediately observable effects of science and the relatively slow functioning of the democratic spirit, and due partly to the selective attitude and hence support from the authorities, Mr. S alone (with a strong emphasis on technology) gradually narrowed and subsequently occupied the whole scope of the critical movement of “learning from the West.”

We may or may not agree on the exact reason why scientism came to prevail. That is of little importance in our present discussion. What matters most is that we have no reason to ignore, to keep silent about, or to reinforce, consciously

²³ Here I limit my analysis to the Chinese context. However, according to my personal discussion with Japanese systems scientists, what I have found is equally true in Japan.

²⁴ “China” in Chinese means “The Central Kingdom,” therefore it is just natural that there is *xi-yang* and *dong-yang* to the kingdom.

or unconsciously, that prevalence, for whatever “good” intention, especially at the present time, when our Western counterparts have become critical of it in their own societies.

It seems apparent that recent systems research and practice in the East have not been able to escape from the influence of scientism. In our systems community, academic journals and research papers are expected to contain, and preferably be full of, mathematical equations and models. Among research projects, those which are technology-oriented predominate and are well funded. Should human factors be undeniable, they are generally expressed as secondary variables. What concerns us is not human values or interests, but merely measurable (i.e., quantifiable) human behavior. Checkland’s (1981) methodology, which in the West is seen as critical of scientism, is commonly interpreted as another powerful means to reach predetermined ends or more precisely to solve “ill-structured problems that have been decided (perceived and expected) to be solved” (Li and Zheng, 1995, p. 54). Habermas is condemned partly because he is believed (erroneously) to be “totally rejecting science and technology” (Xie, 1994, p. 31).²⁵ And so on. Few of us as systems scientists are concerned with the modeling of models. Few are interested in studying human relations to search for the reasons for the success or failure of projects, let alone human values and interests for their own sake. Fewer still bother to ask questions such as: As an expert, do I have vision? What does that vision mean? and Does it matter?

We say that Confucianism was preoccupied with feudal moral values and isolated self-cultivation, exclusively and without balance. Now, do we not need to be aware of the real influence and danger from the opposite direction? Hundreds of years ago, Confucianists recognised that there should be different ways to investigate “truth and principles,” “people and events,” as well as “(human) affairs” (recall the message about “many ways for investigating *lis*” quoted from Cheng Yi). In our modern time, do we recognize this? Do we remember this? Are we searching for different approaches to tackle those *lis*? Or are we prepared to do so? Do we believe that those ways or *lis* are merely feudal teaching or Confucian “empty words” and therefore should be out of the question? Or should we simply forget all *lis* except “scientific truths”? We know that the Confucian view of harmony reduced the Universe into human nature or Mind alone; but haven’t we just reduced the Universe to something else?

In this paper I am not seeking to provide ready answers to these questions. Of course, I am seriously concerned with them, because I believe that they are crucial to a *systems* scientist, crucial to any *systems* research and practice. They are not empty words but real and serious challenges. Holding this belief, I now move to study the *lis* of *wu*, *shi* and *ren*, the foundation stones of WSR.

²⁵ Among other similar statements, one reads, “Habermas holds an attitude which totally repudiates the social function of science and technology in contemporary capitalist society. Of course he is fundamentally wrong” (Xie, 1994, p. 31).

4. WULI, SHILI, RENLI

4.1. A Conception of Reality

I begin the elaboration of *wuli*, *shili*, and *renli* by declaring a conception that the reality we are to tackle in systems projects can be viewed as constituted and conditioned by *wu* (objective existence), *shi* (subjective modeling), and *ren* (intersubjective human relations). In my view, the notions of *wu*, *shi*, and *ren* are parallel to the Confucian conception of “the unity of Being, the unity of Existence, the unity of Life, and the unity of Value.” *Wu*, *shi*, and *ren* in systems projects are by no means static, fixed, or isolated. Rather, they are distinguishable yet interwoven manifestations of the dynamic and relational process of *yin* and *yang*. I would like to make it clear that such a conception of reality is based on a belief which we consider most informative and instructive to guide systems research and practice, especially in the Oriental context, rather than capable of being “proven” in any ultimate sense.

According to the Confucian teaching that “all kinds of things and affairs in this world have their own *li*,” we contend that *wu*, *shi*, and *ren* naturally have their inherent *lis*, which are different and distinct from each other, changeable in themselves, and influence and condition each other in every concrete circumstance. Again, following the Confucian argument that there are different ways for investigating different *lis*, we suggest that, in conducting systems projects, we systems scientists should bring *wu*, *shi*, and *ren* into our consideration, and accordingly search for appropriate methods to study *wuli*, *shili*, and *renli* as well as their manifesting interrelations. However, certain *li(s)* may be perceived by participants as more “crucial,” “major,” “dominant,” or “urgent” than others at this or that moment of a given project.

4.2. *Wuli*

Wu, according to [*The*] *Chinese–English Dictionary* (1988, p. 733), means “(1) things, matter; (2) the outside world as distinct from oneself; (3) content, substance.” For our approach, we extend and confine the notion of *wu* to the whole range of existence in the objective world: not only material conditions, but also man-made constructions and arrangements. By “material conditions” we mean factors relevant to systems projects such as natural resources, the physical environment, climate, location and geographic features, population and its structure, transportation, power and communication facilities, available data and financial resources, production capacity and outputs in the past and present, etc. In the group of “man-made constructions and arrangements” we include formal constitutions, organizations, their hierarchies and structures, as well as existing official regulation systems, and so on. Needless to say, in our experience of previous projects, *wu* has been a major aspect of the reality which, on the one hand,

we had to deal with and, on the other hand, conditioned the conduct and fate of the projects.

Accordingly, *wuli* to us are those laws and regularities which underlie, form, and govern objective phenomena, in the sense of Zhu Xi's *lis* for carts and ships, bricks and bamboo chairs. *Wulis* are relatively stable, yet they are changeable along with increasing human cognitive competence. For the same kind of *wu*, the manifestation of its *li* can be different for different concerns in different circumstances. In this regard, Newton's law and Einstein's theory of matter and movement are a good example: both are true, but in different situations and for different purposes, one appears more directly relevant and operationally workable than the other.

In studying *wu* and *wuli*, what we are searching for, to use Fung's (1948) terminology, is "positive knowledge," or, in the terms of Popper (1966), "descriptions of primary facts." Generally, such knowledge should be produced from observation. It should be testable and falsifiable. It should also possess the characteristics of genuineness, authenticity, accuracy, and precision.

In terms of epistemology and methodology, I consider Cheng-Zhu's teaching to be more relevant than that of Lu-Wang here. As we saw in Section 2, Zhi Xi, as Fang (1973) put it, was an epistemological dualist [Popper (1996, Chap. 5) might say that Zhu Xi was a "critical dualist"]. The value of Zhu Xi's teaching, for our purpose, lies in his recognition of a radical difference, not only between mind and matter, but also between the knowledge of sensory perception based on experimental observations and the knowledge of rational thinking accumulated through the application of logic. In his teaching, Zhu Xi tried, although he was successful to only a limited extent, to seek knowledge in and from the external world through a kind of empiricism.

Given the tendency of scientism in the contemporary Eastern context, which may become more dominant due to the current strong desire and heavy pressure for economic development, I do not want to add fuel to the fire by emphasizing further the importance of objective conditions, principles or laws. Rather, I would like to draw attention to the unavoidably value-laden, theory-guided, partial, and uncertain nature of our knowledge of "objective" facts, data, models, criteria, scientific method(s), etc., which have been addressed by Western thinkers from Kant (1929, 1951) to Kuhn (1970) and beyond. I have no desire to discuss this issue in depth here. Rather, to make my point, I just quote some great scientists—Einstein, Heisenberg, von Bertalanffy, and Gray—as food for further thought:

As far as the laws of mathematics refer to reality, they are not certain; and as far as they are certain, they do not refer to reality. (Einstein; cf. Capra, 1992, p. 49)

Every word and concept, clear as it may seem to be, has a limited range of applicability. (Heisenberg; cf. Capra, 1983, p. 33)

Any statement held from a certain viewpoint has only relative validity. (von Bertalanffy, 1968, p. 248)

We must underline the value of general systems theory in terms of its perspectivist qualities—that is, its insistence that no world view, its own included, is ultimate truth or ultimate reality. (Gray *et al.*, 1969, p. 33)

I consider that this aspect can be tackled more directly by searching for *shili* and *renli*, which I now address.

4.3. Shili

Shi as a noun, according to [*The*] *Chinese–English Dictionary* (1988, p. 625), means “(1) affair, business; (2) trouble, accident; (3) job, work; (4) responsibility, involvement.” For our purpose, I consider the notions of affair, involvement, and engagement to be most relevant.

Accordingly, we use *shili* to denote patterns of engagement, as well as the ways we follow so as to conduct systems projects smoothly and successfully. *Shili* does not pertain to “primary facts” but emerges from “real experiences.”

The meaning of *shili* can be further “suggested” (rather than rigorously defined) by the following examples. The first example is about software. I was once asked whether there was a spell-check function in a particular word processor. The questioner asked this question because he wanted to use it but could not find it. Without having used the package myself—therefore without any detailed “know-how” of it—I replied that I thought it was most likely that there would be such a function, because I believed that the manufacturer would need to include a spell-checker in its product if it did not want to lose its market share. My thinking in this case, I consider, is a piece of *shili*. The inclusion of a spell-check function in a word processor is not a “necessary” universal law or “objective” principle, but a subjectively perceived “better” way to do “good” business.

My second example concerns information systems development methodologies. Generally, to develop a workable information system, we may employ structured methodologies such as SSADM (e.g., de Marco, 1979; Cameron, 1989; Yourdon, 1989; CCTA, 1990); we may follow the principle of prototyping; we may adopt newly developed approaches which draw on insights from Checkland’s (1981) Soft Systems Methodology and other similar methodologies (e.g., Stowell, 1995; Wilson, 1984). Which one we pick will be dependent on the particularity we conceive in differently situated development environments. That is another piece of *shili*. We will most probably choose not to buy hardware first, select software next, build conceptual models afterward, and investigate user requirements and the system environment last. Of course, we can choose to do so if we want—but what for? Again, all those methodologies are not “natural” laws, or “objective” principles, no matter how well they are empirically tested.

They are only beliefs which we consider probably have some value for carrying out our affairs in “better” ways.

In the above examples, and in other situations, *shi* is not constituted by “facts,” nor is *shili* “objective.” Please note that in the above description, I used words such as believe, think, would, want, desire, can, may, choose to, better, etc. All these words have meanings of “suggestiveness” rather than “necessity.” I choose to use these words because *shili* to me is somehow subjective, uncertain, and varying, compared with *wuli*.

However, *shili* can also be considered as “necessary,” since to conduct systems projects in a rational (i.e., organized) way, *shi* should be our concern and “good” *shili* had better be obtained and followed. For certain reasons, although so far I cannot imagine any, we could choose not to study or follow *shili*. Yet mere common sense will be enough to foresee what would happen if we choose such a path. With good knowledge of *wuli*, but without knowing *shili*, our projects can still turn into messes.

I consider that *shili* should also cover ways of seeing; that is, our modeling of models (van Gigh, 1993). This aspect of *shili* may have particular significance at present. Due partly to the influence of scientism, most of us as systems scientists in the East, and I believe in the West as well, tend to take our models as given; we tend to believe that, for any of our “systems” concepts, there must exist something correspondingly concrete. Rather than taking “system” as an epistemological device for understanding some properties or patterns of the complexity which concerns us, we tend to take “system” as objective existence only, claiming that “systems exist everywhere in nature and in human society. For example, the solar systems, the human body, a family, and a manufacturing enterprise are all systems.” In saying this, we forget that a system is a map rather than the territory. In my view, without a good way of seeing, there cannot be a good way of doing. If we agree that a way of seeing inherently implies or suggests a particular way of doing (Romm, 1995), both ways of seeing and doing need to be covered by *shili* in WSR.

Shili, as subjective suggestions or appropriate ways we had better follow to organize systems research and practice, can take various forms. It can take the form of common sense, illustrated by the saying “seeing is believing.” In this form, quite a lot of *shili* has been summarized in the military as well as the management literature; for example, the principle of “Those who know themselves and their enemies well will win,” and many others, in *The Art of War*, an ancient Chinese classic. *Shili* may also take the form of more formal methodologies or guidelines, for example, SSM (Checkland, 1981; Checkland and Scholes, 1990) and TSI (Flood 1995; Flood and Jackson, 1991b).

Since *shili* is neither “objective laws” nor “natural principles,” but “subjective suggestions” or “common sense,” unlike *wuli* it cannot be proven, tested, or falsified. With *wuli*, we can make *wus* work. With *shili*, despite following a well-tried

procedure, we may still encounter surprises and unexpected outcomes. Therefore, I do not think that knowledge of *shili* can be obtained in the same way as we study *wuli*. We need more flexible approaches. As a preliminary suggestion, I consider that methods within the spheres of reflection, intuition, discussion, mutual-informing, and hermeneutics may have a significant role to play.

4.4. *Renli*

According to our experience, the tackling of *wuli* and *shili* in systems projects can be effective only when *renli* is recognized and adequately addressed, whether acknowledged or not. In other words, without addressing issues such as “Why use *wuli* for this purpose but not for that one?” and “Why follow this *shili* but not others?” Seen in terms of relationships between stakeholders, there is no hope of conducting a systems project smoothly or successfully.

Simply put, the concept of *ren* is concerned with the intersubjective aspect of human beings. *Renli* to us is therefore manifestations of interactions and encounters among different value systems, interests, motives, and desires, as well as our norms for addressing and handling those interactions and encounters.

Renli in WSR highlights the importance of human relations, which we consider in most circumstances to determine the fate of systems projects. This idea is highly compatible with Confucian teaching, according to which dealing with social affairs is the way toward Man’s full realization and development. As Mencius (4a/2) once put it, “The sage is the acme of human relations.” The importance of human relations in the Confucian concept of *li* is also well illustrated in the following two messages:

Chinese philosophy is primarily directed toward moral and political goals. Even ontological and cosmological speculations are not without moral and political significance. The *li* (principle, reason) in neo-Confucianism for example is a concretely rational ideal and idea. *Li* is not something divorced from Man’s basic life-experience in relation to himself, other men, and things; it is taken to be the basis for achieving social harmony and administering political order. (Cheng, 1972, p. 161)

... [T]he Way is the way of human relations. This is of course the traditional Confucian concept. As the Neo-Confucianists said, it is the way according to which the ruler, the minister, the father, or the son should behave. As the Ch’eng brothers said, “The relationship between father and son lies in affection. That between the ruler and ministers lies in respect. From these to the relations between husband and wife and between the elder and the young, all that is done should be in accordance with *Tao*. (Chan, 1967, p. 28)²⁶

Such tradition is no doubt a very deep-seated and strong one in Oriental cultures. Compared to their Western counterparts, the Eastern (especially the Chinese) societies are more human relation oriented than law regulated. It is no

²⁶In this quotation, *tao* has the same meaning as *li*.

secret that in contemporary China, if one has trouble, the first reaction is not to appeal to formal laws or regulative procedures, but to look for *guanxi* (favorable informal personal connections).

Thus, it is not difficult for the East to realize the necessity of tackling *renli* in systems projects or in more general situations. A good example at the present time is that, in spite of great pressure for economic development from the Government, as well as from the grass-root sectors, scholars in the East insist on claiming that the next century will be a *Ren Xue* (The Learning of Human Nature, or, “Humanology”) one—although their idea of what *Ren Xue* ought to be should be open to question (see, e.g., Chen, 1995; Feng *et al.*, 1994). In fact, much has been said about human motivation, human behavior, and human relations in the contemporary Chinese literature on behavioral science and “soft science”—particularly since the translation of Checkland’s (1981) book *Systems Thinking, Systems Practice* into Chinese.

But, and the word “but” should be in capital letters, what is our *purpose* of dealing with *renli*? What *aspect* of *renli* occupies our attention? What *approaches* should we use to investigate and tackle *renli*? These questions, which I consider crucial, basically remain unexplored or even beyond our research horizon—however, see Midgley *et al.* (2000), in this special issue, for further thoughts on them. In the following, due to limited space, I address only two relevant issues which, as far as I am aware, have not surfaced previously in the Eastern systems literature: (1) what our long-term purpose should be in tackling *renli*, and (2) how to overcome the dark side of Confucianism in this regard.

First, I argue that tackling *renli* is not an end, but a means. By this I am not saying that we should manipulate human relations in an instrumental way for realizing whatever predetermined ends we might hold. In sharp contrast, I believe that human values and their interests, however different and even conflicting they appear to be, should be studied and served for their own sake, rather than being repressed in the Confucianist name of *tienli* (the law of Heaven), or in the modern interests of “the majority in society,” “scientific truth,” “a prerequisite for stability and development,” or whatever.

What I mean by “tackling *renli* is not an end” is that our long-term efforts in dealing with human relations should be directed to foster and to cultivate a culture of rational and formal regulation, a culture currently weak and urgently needed in the Eastern context. In that context, the other side of “the importance of human relations” is the ignorance or handicapping of rational systematic constraints in social affairs. Why do we have to look for *guanxi* rather than appealing to laws? Because we lack a culture of giving respect to, and following, formal laws and regulation systems.

As I have illustrated in the previous section, the dark side of “The Unity of Heaven and Man” was usually used by the powerful to legitimate whatever they wanted to do or whatever happened to exist. Compared with the “free will” of

individuals, or groups of individuals, laws and formal regulation systems lose their importance and hence become empty words (here the individuals need not necessarily be authorities or officials; they can be anyone who has the “right” to interpret and to do things in whatever ways s/he likes). In a game without effective rules, actions might be rational for the system part(s) in the short term, but their effects are bound to be irrational for the whole in the long run.

As systems scientists, in conducting local and temporal concretely situated systems projects, we have to care for *renli*, to address and tackle them skillfully. There is no doubt about this. But, at the same time, we should recognize the traditional dark side of human relations in the Eastern context, and try our best to overcome this dark side through our skillful systems practice. Without a conscious and critical reflection on *renli*, successes in individual projects will reinforce the unintended “dark side” and will kill an emerging culture which respects “ruling by law” rather than “ruling by person.”²⁷ It is in this sense that to deal with *renli* is ultimately to *reduce its importance*.

To overcome the dark side, I suggest moving beyond the Confucian doctrine of isolated self-cultivation toward a paradigm of communicative intersubjectivity. As I have illustrated, emphasizing merely self-cultivation has constrained the Confucian ideal of investigating and following *renli* within an isolated subjective consciousness. It has blocked Confucianists from addressing and reaching truly ethical morality and human relations, their sincere ethical ideal always ending up converted into autocracy and conservatism. Without adequately realizing and tackling this aspect of *renli*, any efforts to deal with human relations will not be truly ethical but merely instrumental.

We have already seen in contemporary systems practice manifestations of the dark side of the Confucian teaching concerned with *renli*.²⁸ We as systems scientists tend to research human behavior rather than people’s values and interests, because we tend to accept values and interests as given from somewhere, without any questioning. We are keen on acquiring and integrating “wisdom” from “experts” in “different scientific fields,” but ignore the rich experiences and infinite creativity of various social sectors, from those “full of the street,” because even if we agree with such ideas as “systems *of* people” and “systems *for* people” (let us suppose so for a moment), we are hardly aware that if a system is not *by* people, it cannot be truly *of* and *for* them. We tend to think that human values and interests are the problems of sociologists, the business of authorities, not the concern of us as scientists, because our tenet is still “Give me the sys-

²⁷ For similar arguments, but with a focus on the relationship between Chinese culture and economic development, see, for example, Fan (1995).

²⁸ Against a broader background, some have begun to argue that the recent (post-1997) economic and social turmoil in post-Confucian and Asian societies can be explained mainly in terms of just this kind of *guanxi*, i.e., inappropriate relations among big enterprises, the banking-financial system, and the government.

tem and I will give you the solution.” However, in doing so, haven’t we reduced the whole into a particular part(s)? When we take human values and interests as given, when we concentrate on nonhuman “facts” alone, when we degrade human desires into secondary variables or exclude them from our model, can we claim any longer, in any serious sense, that we are *systems* scientists?

In this section I have presented the ideas of *wuli*, *shili*, and *renli* individually. However, I need to repeat that they actually constitute a differentiable, larger whole, and at the same time condition and shape each other. As a differentiable whole, they should not be reduced into any particular *li*. On the one hand, since they are distinct and heterogeneous, each *li* should be investigated and tackled with different approaches. On the other hand, since they are interwoven with each other, no single *li* should be studied in isolation from the investigation of the interplay among multiple *lis*. For this, both the West and the East have much to offer. In the West, we have Plato’s “oneness and manyness,” Ackoff’s (1981) “synthesis and analysis,” etc. In the East, in Confucian teaching, we have, on the one hand, Cheng Yi’s saying that “the *li* of a thing is one with the *Li* of all things” (*Yi Shu*, 2A/1a) and, on the other hand, his suggestion that there are “many ways of investigating *li* to the utmost.” In particular, isn’t Cheng Yi’s “truth and principles” our *wuli*? Doesn’t his way of “talking about people and events” have relevance to our *shili*? Shouldn’t we, with regard to *renli*, pay more attention to the manifestation and handling of interactions among human beings, as Cheng Yi suggested?

5. CONCLUSION

In conclusion I shall not repeat what I have presented in this paper. Rather, I would like to surface two previously unexplored points.

First, WSR itself is a piece of *shili*, in the sense that WSR is a suggestion for seeing and doing things—in our case, to improve and to popularize systems research and practice through the effort of pursuing appropriate methods to address and tackle differentiable *wuli*, *shili*, and *renli* in a systemically informed way. It is subjective because it is based on our personal understanding of Confucian teaching as well as on contemporary systems science. It is also subjective because it is built on our interpretations of previous systems projects which necessarily constitute limited experiences. Yet it also appears to me that, in employing WSR, we are able to explain most successes and failures in systems projects in the Eastern context. Whether WSR has any longer-term normative or prescriptive significance for improving systems research and practice remains to be seen, and can be assessed only through reflections on further applications.

My second point is that, if we are to practice WSR, we will be engaging in two encounters. One is with the exclusive preoccupation of isolated self-cultivation in the conventional Confucian teaching, and the other is the contempo-

rary tendency toward scientism. While the dark side of Confucianism imprisons subjective creativity within the “killing value curtain,” contemporary scientism turns everything into objects, including the subject. To me, both are enemies of systems research and practice.

Thus my conclusion: Although WSR is not “the only feasible and effective way” for sociotechnical systems design and management, research and practice, it might not be a bad idea to pursue.

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